

# SRS Implementation Summary Report

**Project:** Full-Stack Typing Speed Game Application | **Ref:** Burdenoff Intern Take-Home  
**Date:** August 2026 | **Status:** Approved & Fully Implemented

## 1. Executive Architecture Summary

This summary report details the full-stack architectural implementation of the **Typing Speed Game Application**. The solution comprises a Next.js (TypeScript) single-page client, a Bun + GraphQL Yoga API gateway, Prisma ORM with PostgreSQL relational persistence, and containerized Docker Compose orchestration.

## 2. SRS Traceability Matrix

| SRS Section & Requirement                                                                                             | Implemented Feature & Architecture Details                                                                                                                 | Status   |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>3.1 User Registration &amp; Auth</b><br>Unique username/email, salted bcrypt, JWT sessions, protected GraphQL API. | JWT auth middleware with bcrypt password hashing in GraphQL Yoga resolvers (`register`, `login`, `me`).                                                    | COMPLIES |
| <b>3.2 Typing Mechanics</b><br>20 random alphabets, real-time timer at 0s, focus lock, 0.5s error penalty.            | Implemented `TypingEngine.tsx` with programmatic focus locking, 20 random alphabets [a-z], real-time timer, and immediate +0.5s error penalty calculation. | COMPLIES |
| <b>3.3 Scoring &amp; Leaderboard</b><br>Final score = Raw + Errors x 0.5s, LocalStorage PB cache, global leaderboard. | Local storage personal best caching + server-side GraphQL `saveGameResult` and `getLeaderboard` sorted by total score.                                     | COMPLIES |
| <b>4 &amp; 5 Tech Stack &amp; Database</b><br>React/Next.js SPA, Bun, GraphQL Yoga, PostgreSQL 16 + Prisma ORM.       | Prisma schema defining `User` and `GameResult` 1-to-N entities with cascade deletion; Dockerized Bun backend runtime.                                      | COMPLIES |
| <b>7 QA &amp; Testing Strategy</b><br>Frontend keypress & arithmetic tests, backend score calculation & auth tests.   | Automated Vitest unit test suites for frontend sequence/scoring math and backend JWT/penalty arithmetic.                                                   | COMPLIES |
| <b>8 Infrastructure &amp; Setup</b><br>Docker Compose with postgres:16-alpine and backend containers.                 | Fully configured `docker-compose.yml` orchestrating PostgreSQL on port 5432 and backend on port 4000.                                                      | COMPLIES |

## 3. Highlights & Key Features

- **Focus Lock Engine:** Programmatically locks focus without requiring manual mouse interaction.
- **Penalty Calculation Integrity:** +0.5s per error added to final duration, validated on client & backend.
- **Dual Score Persistence:** Instant LocalStorage PB caching + async GraphQL PostgreSQL persistence.
- **UI Aesthetics:** Vibrant dark mode with glowing glassmorphism cards and neon status badges.